

Architectural Particular Specification

ALUMINUM WINDOWS, GLASS CANOPY, ALUMINUM LOUVRES & FEATURES

SECTION 9.1

Rev	Date	Description
0	14 APR 2022	First Issue
A	13 MAY 2022	Tender Addendum No.2

SECTION 9.2

ALUMINIUM WINDOWS, GLASS CANOPY, ALUMINIUM LOUVRES & FEATURES

PART 1 GENERAL

1.01 SUMMARY OF WORK

- A. The systems shall include, but not limited to the following, and at those areas as indicated by the drawings by the Architect:
1. Aluminium Windows & Glass Doors
 2. Aluminium Louvres (fixed and demountable)
 3. Aluminium Features and Claddings
 4. Glass Canopy
- B. The scope of works shall be defined by the architect's drawings, specifications and other documents and instructions which pertain to the works.

The Contractor shall design, engineer, test, fabricate, supply, deliver, install, and guarantee all construction necessary to provide a complete system of Aluminium Windows / Glass Doors / Louvres / Aluminium Features and Claddings / Glass Canopy / Skylight including any measures that may be required to that end, notwithstanding any omissions or inadequacies of the design intent drawings and/or specifications. Tender sum to be itemized and broken down, with rates entered in the Schedule of Rates. The prices therein will be used as the basis for pricing any additional or omitted variation works.

- C. The works shall include, but not limited to:
1. Aluminium Features and Claddings / Glass Canopy
 - a. Design, fabricate, supply, delivery and installation of all aluminum windows wall, aluminum features, glass canopy, skylight work as specified or shown on Drawings.
 - b. Aluminium mullions and transoms / stainless steel framing members, including capping to transom and mullion members as shown on drawings.
 - c. Steel structural frame and reinforcement such as galvanized mild steel sections, G.M.S. hanger frame / system for supporting such window wall at the top, and other steel reinforcement required for a complete installation (whether these structural frame / reinforcement are illustrated on drawings or not).
 - d. Structural Glass and Glazing Material for Window Wall as specified below and ASD General Specification edition 2012.
 - e. Fixings, connections and provision of anchorages. All Aluminium / Steel Supporting Frames should be anchored to the concrete structure. Provide and install cast-in embedment / brackets in concrete.
 - f. Provision of safety devices, including but not limited to stainless steel safety chains, fasteners, brackets or the likes, for each module demountable louvers and decorative grilles.
 - g. Glass and glazing materials including glass spacers, gaskets, sealants (including structural sealant), backer rod, Norton tape etc.
 - h. Spandrels, dressings to roofs/abutments, integral louvres and screens to intakes/exhausts for MVAC system.

- i. Aluminium copings at top of parapets & external cladding where shown on drawings;
- j. Ironmongery and fittings.
- k. Sills, copings, water bars, flashings, seals and other attachments.
- l. All reinforcement and stiffeners for the aluminum cladding system required for the installation.
- m. Fire seal / Insulation of requisite fire resistance performance against fire spread between storeys / designated compartments.
- n. Isolation of dissimilar metals and moving parts.
- o. All preparatory work to the adjoining structure for attachment of the works.
- p. Sealants completed with appropriate accessories such as backer rod and other filling materials, joint fillers, etc. within work of this section and at boundaries / interfaces with work of other sections.
- q. Flashing and sealing of surround to aluminum frames by waterproof grout, slurry coat, sealant, metal flashing, etc.
- r. Thermal insulation at non-vision areas, spandrel, etc.
- s. Test mock-ups and corresponding tests (including performance testings as required by the Independent Checker under relevant requirements under the Buildings Ordinance and Regulations, and other relevant authorities) carried out by an independent laboratory approved by the Architect and to the satisfaction of standards listed in this specification.
- t. 100% Field water tests to window frame grouting for demonstrating their resistance to water leakage conforming to the standards listed in this specification.
- u. Material tests by recognized laboratories.
- v. Final cleaning and appropriate protection to the installed window system against accidental damages during the course of construction works and prior to handover to the Employer all to the satisfaction of the Architect.
- w. Proposed method and means of protection to be submitted for Architect's Approval.
- x. Lightning protection and means for connection to earthing terminals of the development.

2. Aluminium Windows, Glass Doors, Louvres and Glazing Systems

The works include, but are not limit to :-

- a. Design, engineer, fabricate, supply, delivery and installation of all aluminum windows, doors, louvres and glazing systems as specified or shown on Drawings.
- b. Lightning protection and means for connection to earthing terminals of the development. For aluminum window units to be installed in precast concrete facades, an earthing lug casted into the precast façade is to be provided and connected to the aluminum window unit. Ensure the other termination point of the earthing lug will protrude beyond the precast façade at approved location to the satisfaction of the Architect for onward connection to the nearest earthing terminal by the Electrical Installation NSC.
- c. Supply and installation of complete fittings, ironmongery, hinges, fixing screws and lugs, reinforcement, bearings, gaskets, sealant (including structural sealant), capping and other necessary accessories.
- d. Supply and installation of glass and structural glazing.
- e. Mastic compound and filling transom and mullion couplings.

- f. Flashing and sealing of surround to windows, louvres, doors by waterproof cement grouting and slurry coat.
 - g. Silicone pointing and caulking to surround of windows, louvres, doors or as otherwise specified.
 - h. The setting up, testing and test results of mock-up sample windows selected by the Architect and of those window units / systems requiring performance testings by the Independent Checker under relevant requirements under the Buildings Ordinance and Regulations, and other relevant authorities.
 - i. Testings upon completion, including 100% field water tests to window frame grouting (prior to application of external finishings) as well as the aluminum window unit itself for demonstrating resistance to water leakage.
 - j. Forming of holes on aluminum backing plate behind aluminum louvres for receiving air ducts, installation of exhaust fans, etc. including coordination works with the other relevant sub-contractors on locations, sizes and work sequences of the same.
 - k. For sliding doors accessing to balconies of accommodation units, in addition to the lockset / handle for normal operation of the doorset, a "child-proof lock" shall also be incorporated as generally indicated at Appendix "I" of this specification.
 - l. Hardware for sliding doors accessing to balconies of accommodation units for persons with a disability shall be lift and slide with a barrier free threshold of not more than 20mm.
3. Apart from the above, the scope of works shall also include, without being limited to:
- a. All stainless steel window washer tiebacks as shown in the tender drawings or as required to complete the respective system installation.
 - b. All fire protection, including all necessary fire proofing materials applied on all steel members of F.R.R. fixed windows and any necessary fire protection measures to the Works to the satisfaction of the Independent Checker under relevant requirements under the Buildings Ordinance and Regulations, and Fire Services Department.
 - c. All aluminum cover panels and cappings where indicated.
 - d. All caulking, sealing & flashing including sealing, backing rods and joint filler, etc. at junction with roof waterproofing, external wall tiling, & other areas in contact with the external concrete wall.
 - e. Thermal insulation.
 - f. Isolation of dissimilar metals and moving parts.
 - g. Continuity of various metal parts (whether fixed or movable) to facilitate earth bonding and link up with the earthing terminal of the building.
 - h. All bolts, angles, fasteners, anchors, inserts and channels in concrete structure for the anchorage of all works under this section.
 - i. Aluminium cladding / capping and window wall framework finishing, and painting of all exposed steel works under this section.
 - j. Waterproofing groutings and coatings at interfaces of windows frames and concrete openings as indicated on drawings.
 - k. Provision of valid fire test reports and certificates to Architect's and relevant authorities' satisfaction issued by independent laboratories accepted by the Independent Checker under relevant requirements of the Buildings Ordinance and Regulations, or other relevant authorities for the fire resisting and insulation performance of all fire resisting windows / window walls / glass walls, etc.

- l. Provision of hinge restrictors to all window installations of accommodation units.
 - m. Provision of strainer to hold the window in place in case of typhoon.
4. The main contractor shall also be responsible for:
- a. The design, engineering, fabrication, supply, transportation to site / pre-cast facade factory within 800km from the site as the case may be, installation and testings of the whole works including grouting behind frames, caulking, glazing, pointing etc. in accordance with the tender documents.
 - b. Save where explicitly excluded, provide and install all components, fixings and accessories etc. required to complete the system as is intended, regardless of whether such have been explicitly described or not.
 - c. Shop drawings, engineering proposal, structural calculations, engineering data, material certified by the manufacturer/supplier for the installation including the element, glass panels, the supporting frame, fixing and anchorage of the system to the building structure, test reports endorsed by an independent testing laboratory upon completion of performance tests as required by the Buildings Department, structural sealant compatibility tests to be performed by an independent laboratory to be approved by the Architect / relevant authorities, impact load tests to tempered glazings to demonstrate their capability in withstanding the designed wind load and / or the horizontal imposed loads as referred to in Table 3 of Building (Construction) Regulations, Code of Practice on Wind Effects in Hong Kong 2004 including the consideration of artificial base level, edge zone effect, and all other relevant documents and data of the window framing, sealant, and glazings for glass wall, window wall, glass balustrades / railing and aluminum window unit, bi-fold doors are to be endorsed by a Registered Structural Engineer acceptable to the Architect for submission to Buildings Department for obtaining necessary approvals and consent for his works. The main contractor shall bear the consequences of any delay due to the failure in obtaining the necessary government approvals and consent.
 - d. All calculations must conform to Codes and Regulations presently being enforced in Hong Kong. The Architect may require the Contractor to alter details of fixing and anchorage in order to satisfy Building Regulations. The design and drawings shall be checked and endorsed by an experienced qualified structural engineer (RPE or RSE) engaged by the contractor prior to submission to the Architect/Engineer. Allow for the processing time for the approval and consent.
 - e. Design and provide all connections, brackets and fixings by approved methods so that all loads can be transferred from the glazing system to the base structure in accordance with the design criteria and in a manner that prevents excessive joint displacement, slippage or distortion.
 - f. All components are to be designed for the maximum tolerance of the system, and due consideration shall be given to additional forces from prying action and bolt group effects.
 - g. There shall be no direct fixing or close contact between any part of the glazing system and the base structure or internal linings, except through approved bracket connections.
 - h. Brackets and related components shall be scheduled and described in detail on shop drawings. Show details of all related components and connections to areas by others.
 - i. Where additional supporting frame, fins and/or bracing etc. is needed to support the aluminum claddings / features / louvers etc as shown in the drawings, the frame, fin, bracing etc. shall be designed, installed and provided by the contractor.
 - j. Scheduling and monitoring of the work,

- k. Samples, full size unit mock-ups and test units,
 - l. Performance testing of the window wall framing and glazing assembly, cladding assembly, louvre assembly, and aluminum window framing and glazing assembly together with corresponding test reports endorsed by an independent testing laboratory to be approved by the Architect;
 - m. Co-ordination work with the Main Contractor on any cast in items or other builder's works and other trades with work interfacing.
 - n. Protection of materials and works, unit handing-over to the Employer all to the satisfaction of the Architect. Proposed method / means of protection to be submitted for approval of the Architect.
 - o. Guarantees and warranties,
 - p. All final exterior and interior cleaning of the Aluminium Windows / Doors / Louvers and Aluminium Features works, etc.
 - q. Provide all hoisting and temporary site services for proper execution of the works.
 - r. Cost of all delivery & hoisting of material.
 - s. Design, supply and install the window system's earthing / equipotential bonding in accordance with I.E.E. regulations and provide a termination point for the Electrical Sub-contractor to connect and integrate it into that of the rest of the building.
 - t. Make good and replace any damaged parts.
 - u. Provide spare and tools prior to Practical Completion of the Main Contract:
 - v. -1% of each type of window and door ironmongeries
 - w. -1% of each type of glass unit and size
 - x. Coordinate with the building services engineer to ensure that the required overall thermal transmission values (OTTV) are achieved.
5. Responsible for the watertightness of the whole Aluminum Window / Door / Louvres / Claddings, Window Wall System, Glass & Metal Balustrade / Railings/ Bi-fold Doors and its structural stability. Any defect or leakage found within the Guarantee Period shall be sealed and made good all at the expense of the main contractor.
 6. 100% factory water testings to all aluminum window / door / window wall assemblies, together with the grouting around window frame for precast panels to standards listed in the specification, to be witnessed by representative of both the Main Contractor and the Aluminum Window Domestic Sub-contractor at the manufacturing plant before the delivery to the precast panel factory.
 7. 100% field water testings to all aluminum window / door / window wall assemblies, together with the grouting around window frame for precast panels to be witnessed by Architect's representatives and representative of both the Main Contractor and the Aluminum Window Domestic Sub-contractor in accordance with PNAP APP116 and standards listed the specification, are required for all works under this contract.
 8. This specification and the architectural drawings showing the external cladding / capping, window wall, glazing panels, glass & metal balustrades / railings, aluminum features works and window / door / louvre units are for the sole purpose of defining the design intent and performance requirements only. To avoid any misunderstanding or lack of interpretation, the main contractor is hereby advised that the responsibility for the works is totally his, including but not limited to the design, structural calculations, waterproofing performances, carrying out of performance testing and preparation of corresponding reports to be endorsed by an independent testing laboratory acceptable to relevant government authorities and the Architect, shop drawings, material installation,

guarantees, and related documentation and certification as required by the Buildings Department and other relevant authorities, and those stipulated under the Buildings Ordinance (Cap. 123), its subsidiary legislations, and/or other enactments as stipulated in this specification.

9. Schedule of fabrication, delivery of bolts, anchors, brackets, holes, slots, etc. and installation including insertion in concrete structures to meet the construction programme. Co-ordinate sizes of fabricated window wall, glazing panels, cladding, aluminum features, and window / door / louvre units with glass and glazing unit sizes and panel sizes.
10. Manufacturers and erectors shall have proven achievement and experience in similar work for at least ten (10) years and are subject to endorsement by the Architect.
11. Drill-in anchor under tension shall not be used for cantilevered structure.

12. Related Works

Works related to, but is not limited to the following:-

- a. Reinforced concrete structure and external tiling works;
- b. External and Internal Stone Installation works / finishing.
- c. Floor coverings and ceiling construction;
- d. Mechanical and electrical systems (except lightning protection and equipotential bonding system);
- e. Waterproofing (other than flashings, water-proof grouting and slurry paint as defined in this specification and as shown on drawings);
- f. Roofing and roof finishes;
- g. External paving & asphalt works;

13. Measurements

All true measurements as shown on the tender drawings are in the metric system. Responsible for verifying all the dimensions and actual conditions on site.

14. Tender Drawings and Specifications

- a. The tender drawings are diagrammatic and do not purport to identify or solve completely the problems of thermal or structural movements, watertightness, pressure equalization, vapour barriers, fixture and anchorage, flatness and stability of facing and moisture disposal.
- b. The tender drawings do not purport to solve problems at the glass line associated with glass movement, pressure fracture or thermal shock.
- c. The tender drawings indicate profile and configuration required together with relationship to structural frame and building interior elements only.
- d. The tender drawings contain details which suggest directions for solving some of the major design requirements and the contractor may use these details and develop them as he deems fit.
- e. Specifications are "Performance" type and include the minimum requirements of the Window Wall, Aluminium Window / Glass Door / Louvres / Cladding, Glass & Metal Balustrade / Railing, Aluminium Features, Trellis, Bi-fold doors etc. without limiting the main contractor to the method of achieving such performance.

15. Lightning Protection

- a. Incorporate lightning protection device for all the window, cladding, aluminum features, window / door / louvre systems. Install all necessary conductor, wiring connection etc. for these systems and the electrical installation nominated sub-contractor will be responsible for constructing the

earthing system from the lowest and top point of the said systems to ground level.

- b. The metal structure of the above systems shall effectively bonded together as required in the latest edition of the IEE Regulations and shall be incorporated into the building main earthing system.
- c. Proposal on the lightning protection system shall be endorsed by a qualified lightning protection engineer. The whole of the system shall be tested to ensure its compliance with requirements as laid down in the British Standard Code of Practice CP 326:1965 at the cost of the contractor.

16. Development

Materials and methods of engineering development other than indicated or implied by the drawings and specification may be employed when such materials and methods conform to all of the specified performance criteria, the design intent, the design requirements and the applicable codes and standard as specified in the contract, subject always to the acceptance by the Buildings Department and the Architect at his absolute discretion.

1.02 REFERENCE

Applicable Standards

- A. Only the latest editions of the referenced codes and standards as of date of tender shall apply.
- B. The standards, codes, and specifications referenced below and elsewhere in these specifications shall apply; however, similar codes or standards used in the home country of the manufacturer may be used if applicable, subject to acceptance of the relevant authorities and the Architect. In such case the manufacturer shall submit documentation satisfactory to the Architect that such code is essentially equal to the one specified.
 - 1. Specifications for the Design Fabrication and Erection of Structural steel for Building issued by the American Institute of Steel Construction (AISC).
 - 2. The Code for Welding in Building Construction issued by the American Welding Society.
 - 3. Specification for Structure of Aluminium Alloys issued by Aluminium Association (AA).
 - 4. Stainless Steel Designation issued by American Iron and Steel Institute (AISI).
 - 5. The specified documents of the American Society for Testing and Materials (ASTM) including but not limited to:
 - a. ASTM E283- 04(2012) Standard Test Method for Determining Rate of Air Leakage Through Exterior Windows, Curtain Walls, and Doors Under Specified Pressure Differences Across the Specimen
 - b. ASTM E331 - 00(2009) Standard Test Method for Water Penetration of Exterior Windows, Skylights, Doors, and Curtain Walls by Uniform Static Air Pressure Difference
 - c. ASTM E330 - 02(2010) Standard Test Method for Structural Performance of Exterior Windows, Doors, Skylights and Curtain Walls by Uniform Static Air Pressure Difference
 - d. ASTM E1105 - 00(2008) Standard Test Method for Field Determination of Water Penetration of Installed Exterior Windows, Skylights, Doors, and Curtain Walls, by Uniform or Cyclic Static Air Pressure Difference
 - e. ASTM E783 - 02(2010) Standard Test Method for Field Measurement of Air Leakage Through Installed Exterior Windows and Doors

- f. ASTM E2359 - 06 Standard Test Method for Field Pull Testing of an In-Place Exterior Insulation and Finish System Clad Wall Assembly
6. The specified documents of the International Organisation for Standardization (ISO).
7. Japan Industrial Standard (JIS).
8. Specification for Metal Window Wall by the American Aluminium Manufacturers Association (AAMA) including but not limited to:
 - a. AAMA 501.1, Standard Test Method for Water Penetration of Windows, Curtain Walls and Doors Using Dynamic Pressure
 - b. AAMA 501.2 "Quality Assurance and Diagnostic Water Leakage Field Check of Installed Storefronts, Curtain Walls, and Sloped Glazing Systems"
9. Specification for Metal Window Wall by Japan Window Wall Manufacturers. (JCWA).

All current local legislations, regulations, Codes of Practice and Practice Notes for Authorised Persons / Registered Structural Engineer, Practice Notes for Registered Contractors under the Buildings Ordinance (Cap. 123), Buildings Department, Hong Kong.

The contractor should strictly observe and comply with the requirements laid down in, inter alia, PNAP APP 13, APP 16, APP 37, APP 130, ADM 10, ADM 13, APP 110, APP 116 (which shall not be deemed exhaustive) in the course of their design, fabrication, manufacturing, drawings and documents submission preparation, performance testing of the various assemblies / systems, quality control (both off- and on-site), actual construction on-site, testing / commissioning, maintenance during Defects Liability and Warranty Period, and record plan / documents submission preparation for facilitating the application of Occupation Permit upon works completion for all works under this contract.

10. British Standards Institute (BS)
11. European Standard (BS EN)
12. Standards and Code of Practice for the design of structural steelwork elements shall also refer to the Specification for Structural Steel Works.

- C. Unless otherwise specified, all materials shall be of the best quality of the type specified and consistent in colour finish.

1.03 QUALITY ASSURANCE

(NOT USED)

1.04 PERFORMANCE REQUIREMENTS

A. General

1. All coatings when cured shall be visibly free of flowlines, streaks, sags, blisters or other surface imperfections or defects.
2. All finishes shall be all within the colour range of the accepted samples.
3. Dry film thickness of coatings on exposed surfaces when measured in accordance with ASTM B244-68 shall not be less than 1.60 mils except in channel recesses and internal corners, which should be visually covered.
4. All coatings should be repairable by use of an exterior grade air-drying touch-up material available from the coating manufacturer.
5. All moving parts shall be separated, and coated or treated to ensure proper movement without noise.
6. Salt Spray Resistance

In accordance with ASTM B 117-73, withstand a minimum of 3,000 hours exposure to 5% salt solution at 95% R.H., at 100 degree F. with no more than 1.25mm creepage or loss of adhesion from scribed line or cut edges.

7. Humidity Resistance

In accordance with ASTM D 2247-68 (1973), withstand a minimum of 3,000 hours exposure to 100% R.H. at 100 degree F. with no more than a few blisters, size No. 8 as shown in Fig. 4 ASTM D 714-56 (1970).

8. Abrasion Resistance

Withstand falling sand with an Abrasion Coefficient Value of 65 minimum when evaluated per ASTM D 968-51 test method.

9. Muriatic Acid Resistance-withstand exposure to 10% muriatic acid solution - 15 minute spot test with no blistering, or visual change in appearance.

10. Mortar Resistance

- a. 24 hour pat test at 100% R.H. without gaining adhesion or any visual effect on the painted surface of solid colours. Metaric may show slight stain.
- b. Mortar is 75 grams of building lime (ASTM C207-74) and 225 grams of dry sand plus 100 grams of water to make a soft paste.

11. Detergent Resistance

Withstand immersion in 3% synthetic detergent solution for 72 hours @100 degree F with no loss of adhesion and no visible change.

12. Colour Retention

Withstand maximum colour change of SE Unit (NBS) after 5 years exposure at 45 degree South Florida per Test Method ASTM D 2244-68, PPG 3.8.1.4.

13. Chalk Resistance

Withstand maximum chalk rating of No. 8 for colours and No. 6 for white after 5 years exposure at 45 degree South Florida per Test Method ASTM D 659-44 (1970).

14. Testing Certification from the Coating System Manufacturer

Submit as required by the paint manufacturer licensing agreement.

15. In House Test Report

Submit report on four (4) equally spaced intervals for the duration of production as a back-up to the paint manufacturer test certificates.

B. Sealant Compatibility

1. Sealant shall be compatible with approved finish and shall exhibit no deleterious effects to coatings.
2. The Contractor should consult with his sealant supplier in selection of sealant which will exhibit adequate adhesion to the finished aluminum surface. Panel exhibit of the specific coating to be used should be submitted to the sealant manufacturer for tests and recommendations. Peel adhesion tests as described TT - S - 227E and TT - S - 00230C are suggested.

C. Field Touch-Up and Repair

1. Manufacturers shall supply materials for air dry touch-up for spray or brush application.
2. Field touch-up of painted aluminum is permitted only with written permission from the Architect. Unless touch-up is authorized, replace damaged material with new material.

3. Touch up test sample must be approved by Architect prior to any field touch-up work.

1.05 DESIGN REQUIREMENTS

A. Water Tightness

1. No water infiltration shall occur when subject to the effect of water sprinkler at 5 litres/min./sqm for 10 minutes under the pressure of +1.5 Kpa/sqm for all fixed part, +1.0 Kpa/sqm for openable sash part.
2. Water infiltration, in this Specification, is defined as the appearance of uncontrolled water other than condensation on the roomside face of any part of the wall. Double seal or pressure equalization system shall be provided to maintain this water-tightness.
3. Proper concealed drainage system to be approval of Architect shall be provided to drain out any water that may leak through the external sealing.
4. Field water tests shall be performed on completed portions of the Window, Louvre and Door at the Architect's option.

B. Air-Infiltration

The air infiltration shall not exceed the following when subject to a static pressure drop of 10 kg/sqm (equivalent to 13.13m/sec wind velocity).

1. Fixed windows : 1 cubic metre / hr / sqm
2. Openable window: 0.25 cubic metre / hr per window perimeter

C. Structural Properties

1. The design of the Window, Louvre and Door and take into consideration of the structural properties of this construction.
2. The design wind pressure for Window, Louvre and Door System including glass, openable windows shall follow all current "Practice Note for Authorised Persons and Registered Structural Engineers" together with all current statutory requirements notified by Hong Kong Building Authority.
3. When subject to the maximum wind pressure of the design wind pressure:
 - a. Vertical frames shall not have a deflection exceeding 1/180th of the length between supports or 10mm whichever is the lesser.
 - b. Horizontal frames shall not have a deflection exceeding 1/250th of the length between support or 4mm normal to the plane whichever is lesser.
4. The deflection of cantilever slab and/or beam under maximum designed live load will be 6mm. Under the above condition, the mullion and window members, glass and sealant shall not be damaged and not have any harmful permanent deformation.
5. Deflection at mid-span of glass shall not exceed 15mm upon application of half design wind load.

D. Sound Insulation

1. An insulating glass unit with minimum air gap of 12mm. Sound reduction performance of the Window and Door shall be
 - a. 29 dB(A) minimum for Living Room, Dining Room, Family Room and bathrooms
 - b. 32 dB(A) minimum for Bedrooms

E. Concrete Tolerances

Allowance shall be made for accommodating the permitted tolerances for the construction of the concrete structure. Notwithstanding any other clause in this document the maximum tolerance of finished structural members resulting from all human errors, limitation of materials and formwork, deflection of formwork, and any other factors that may affect the locations, sizes and alignment of finished structural members shall not exceed +10mm in any direction from nominal position.

F. Building Movements

Allowance shall be made for accommodating both horizontal and vertical deflections of the structure due to wind effects, deflection of beams, and the long term effects of concrete creep. The allowance shall be additional to that for concrete tolerance.

G. Wind Effects

The maximum horizontal deflection of the whole building structure under wind load will be 1/500 of total building height and the maximum horizontal deflection between two floors under this condition is 1/500 of the floor height.

H. Beam Deflection

The cantilever beams and long-span beams will continue to undergo deflection during the life of the building because of creep of the concrete structure.

I. Temperature Movements

Make provision in the Window, Louvre and Door design and erection for expansion and contraction resulting from solar heat gain and a seasonal temperature range of 0oC to 40oC and daily temperature range of 7oC and a building interior range of 10oC to 27oC.

J. Lightning Protection

The electrical earthing and lightning protection system for the entire building will be installed by others. Co-ordinate with the electrical contractor and provide the necessary connection terminals to permit such bonding.

K. Weight

The total weight of the Window, Louvre and Door including glass and any reinforced structural steel members shall not be more than 50kg/sqm.

1.06 ENVIRONMENTAL, CLIMATIC & PSYCHOMETRIC CONDITIONS

(NOT USED)

1.07 SUBMITTALS

A. Submission Schedule

1. Immediately after award of the Contract, submit a complete submission schedule (in a format to be approved by the Architect) to the Architect for approval listing all the items requiring approval from the Architect, their programme of submission and expected date for getting approval from Architect for their ordering and fabrication of materials. The list of items shall include the followings:-
 - a. Samples of materials, colour and finishes,

- b. Shop drawings,
- c. Technical data, technical information & catalogue of materials & equipment,
- d. Mock-up construction,
- e. Structural calculations.

The above submission schedule will be treated as an indication of the intended programme for submission & application for approval only.

2. Processing Time

Allow time for submittal review, including time for coordination and re-submittals as follows. Time for review shall commence on Architect's receipt of submittal. No extension of the Contract Time will be authorized because of failure to transmit submittals enough in advance of the Work to permit processing, including re-submittals.

3. Sub-standard Submission Materials

No consequential claim on time and monetary compensation from late approval of Architect on any submission materials required under this specification will be entertained should such late approval be caused by the enduring poor quality or sub-standard submission materials.

4. No comment, or failure to comment on, or any other act or omission by Architect relating to this specification and nothing contained in this specification shall constitute an instruction or otherwise impose upon the Employer or relieve the Contractor wholly or partially of any duty, liability or obligation whatsoever, provided that the Contractor shall comply with instructions issued by the Architect which relate to the production of information as applicable to the contents thereof.

B. Shop Drawings

Provide drawings showing all building materials in place and drawings shall include sufficient elevations, floor plans, sections, full size details and explanatory notes that fully substantiate the various systems. Different types of elements such as fixed panels, openable ventilators, window units, louvers, features, cladding, trellis, glass canopies and bi-fold doors and the like must be identified clearly on the elevation drawings. The types, dimensions, and sectional properties of all structural elements, temporary or permanent fixings, weldings, fire stop details, protection to metals and dissimilar metals must be identified clearly in all details & sections. Details shall be fully drawn (not outlined).

In particular, drawings shall include but not limited to the following information:

- 1. Joinery and internal seals;
- 2. Glass and metal thickness in particular fixing lugs and ties etc.;
- 3. Metal alloy, temper and finish;
- 4. Glass strength, tint, coating, opacity, emissivity, weight and safety backing.
- 5. Fastener alloy, plating, diameter, length, and spacing;
- 6. Glazing materials identification and minimum dimensions of structural silicone glazing, where used;
- 7. Sealant identification by product name;
- 8. Relative layout of walls, beams, columns and slabs with dimensions noted;
- 9. Dimensioned position of glass edge relative to metal daylight;
- 10. Provisions for thermal movements and building movements;
- 11. Location of, and details for, any embedded anchors.

12. Connection and interfacing detail with adjacent / adjoining structures, walls, beams, M/E accessories, architectural features, etc. and all interior / exterior finishing materials.
13. Earthing provision.
14. Any temporary fixings for the works during installation.
15. Instructions and explanatory details for the sequence of installation and the installation of all materials, including the glass and re-glazing procedure.

Full specifications of the materials offered, including product name and location on drawings shall be furnished. Verify at the building site all dimensions shown on Architect's drawings. Six copies and one reproducible print for each of the final approved drawings shall be submitted for the Architect's and Employer's use.

C. Structural Calculation and Submission Drawings

The main contractor shall:-

1. Furnish structural calculation and relevant government submission drawings verified and endorsed by a Registered Structural Engineer approved by the Architect, in connection with the design of the works which are in compliance with all applicable Hong Kong Codes and other statutory enactments.
2. The drawings shall include details of the materials, method of fixing, structure details including erection procedure and calculations to show that the window, louvre & sliding doors and cladding system, curtain and window wall, glass balustrade / railing and aluminum features, trellis with glass canopies, bi-fold doors are of adequate strength and stability.
3. The design calculation shall include a synopsis stating the relevant codes and standards, the design loadings and the design consideration, and relevant calculations that substantiate all glass and window wall types and other systems, and their connection and anchorage satisfy the requirements of strength and stiffness under both the working and the test load conditions.
4. The contractor shall prepare calculations to satisfy the wind load indicated for the curtain / window wall, aluminum window / louvers, grille work, glass balustrade / railing, bi-fold doors and other systems and in accordance with good and practical theory and current design practice and in accordance with the rules of the Aluminium Association, AISC, AISI, ACI, Buildings Ordinance (Cap. 123), its subsidiary regulations, codes of practice and other relevant enactments, including analysis for wind and dead loads on framing members, anchors and concrete inserts. Show section property computations for framing members and submit full-size drawings.
5. These structural calculations and submission drawings must be prepared specific for this project works. Existing test reports, data from other projects, or untried or unproven theories shall not be acceptable. Where a conflict between the various codes, legislations, specifications or agencies exists, that code, legislation, specifications, or agencies which has the most conservative / onerous requirements shall govern.
6. These calculations shall include static and dynamic analysis for wind and dead loads e.g. B.M.U. cradles on framing members, structural connectors, anchors, concrete inserts etc. where applicable. Section property computations should also be submitted. Where there are conflicts between codes, legislation, specification, or agencies the most conservative / onerous requirements shall govern.
7. The verified and endorsed, structural calculations shall be submitted to the Architect for checking, endorsement by the Registered Structural Engineer of the project for submission to Buildings Department. The checking and endorsement

by the Project RSE shall not relieve the responsibility of the main contractor in obtaining on time approval and consent for the works.

8. The Architect / Engineer shall review the structural calculations to insure their conformity to the Contract Documents. Architect / Engineer's approval of structural calculations will not relieve the Contractor from any of the responsibilities and requirements as herein specified.

D. Method Statement and Detailed Programme

1. The method statement and the programme shall be the result of co-ordination with other relevant nominated sub-contractors. The document shall also be incorporated into the master programme, process and later assessment of claims.
2. These documents may be modified from time to time to suit site conditions and shall be used to monitor progress and quality.
3. Should the Contractor consider it necessary to fabricate in bulk before endorsement in order to meet completion date, it is entirely at his own risk and no compensation in time or cost will be granted if the material and / or workmanship is eventually not acceptable to the Architect.

E. Samples

1. Submit for approval three (3) sets of labeled samples mounted on timber boards of each required type, colour, pattern and texture of metal finish, on 300mm long sections of extrusion sections and 300mm x300mm sheet aluminum / stainless steel, etc. as required for the various systems. Samples must show extremes of colour and texture variation. Samples will be reviewed by the Architect for colour and texture only. Compliance with other requirement is the exclusive responsibility of the main contractor. Colour and texture range of production material must conform to approved samples.
2. Submit for approval three (3) sets of range samples for each type of glass to be furnished. Samples must be of the proper thickness with a minimum size of 600mm by 600mm mounted on white board surface.
3. Samples for glass must show extremes of light transmission, reflectance, colour of transmitted light as viewed on a white surface, and mottling of reflective coating. For mottling only, the sample must be tempered. The minimum number of sample (comprising one set) is two (demonstrating high and low extremes for all four properties simultaneously), and the maximum number is eight (demonstrating high and low extremes for each property separately).
4. Selected and approved samples for glass shall have permanent identification markings by the glass supplier. The approved samples shall be retained by the glass supplier. The samples shall be made available at the plant for comparison with production material. Upon request by the Architect, the glass supplier shall transport the samples to the project site for comparison with material in place, at no additional cost to the Employer.
5. Submit for review three (3) sets of samples (where applicable samples shall be 300mm in length) of the following:
 - a. Glazing sealants; (including structural sealants).
 - b. Tapes;
 - c. Setting blocks;
 - d. Side blocks;
 - e. Gaskets;
 - f. Shims;
 - g. Ironmongery;

- h. Sweep and compression seals;
- i. Flashing or any other waterproof devices;
- j. Extruded aluminum sections (including the earthing lug) for the various window systems.
- k. Brackets, channels, bolts and other fixing systems. Colour chart should also be submitted for those elements with ranges of colour available, such as sealant and gasket, for Architect's selection.
- l. All samples are to be clearly labelled and identified. Full technical literature and specifications must be provided with the samples.
- m. Approved samples shall be retained by the architect, for comparison with production materials.
- n. Samples should show fabrication techniques and workmanship of component parts, and the design of accessories and other concealed or exposed auxiliary items before fabrication of the work proceeds.
- o. With respect to aluminum window units, prepare a corner section of fully fabricated window unit completed with openable sash, gasket, sealant, lockset, setting-blocks, shims, and the specified glass unit etc. for Architect's approval.

F. Quality Standards Required

a.	Colour uniformity	AAMA - 605.2 : 1980
b.	Film thickness	AAMA - 605.2 and ASTM B224-68 Measurement - 1.6 mil
c.	Specular gloss; 25-35	AAMA - 605.2 and ASTM D523-67 (@60° Viewing angle)
d.	Pencil hardness	F-2H ASTM D523-67
e.	Post formability, 180° bend around 3mm mandrel	ASTM D 1737-62
f.	Film adhesion	AAMA - 605.2
g.	Abrasion resistance; 65 min	ASTM D 968-51
h.	Chemical resistance	AAMA - 605.2
i.	Corrosion resistance	AAMA - 605.2
j.	Weatherometer 500 hr. exposure	Fed. Test Method 6152 Fed Stds. 141a.
k.	Humidity 3000 hrs. Exposure, @100% RH	ASTM D714-56 ASTM B2247-68
l.	Salt Spray 3000 hrs. In salt fog at 95°F	ASTM D714-56 ASTM B117-64

1.08 PRE-INSTALLATION CONFERENCE

(NOT USED)

1.09 DELIVERY STORAGE AND HANDLING

A. Planning of Delivery

Submit preliminary planning reports of the whole stage of the delivery from the factory to the site or to the precast façade factory paying careful attention to the traffic and site condition. The planning shall include packing, storage, protection and unpacking.

B. Product Handling

1. Package and store materials in a manner that will prevent surface damage or contamination, distortion, breakage of structural weakening.
2. Replace any materials damage during manufacture, shipping, storage or erection.
3. Protect materials in place from contamination and damage.

1.10 MOCK-UPS

This part shall be read in conjunction with Clause 8.03 of Specification Preliminaries for the requirements for prototypes and trial areas.

1.11 TESTING

A. Performance Testing Particulars

Submit a procedure plan for the performance test of the various systems, including curtain / window wall, aluminum window, bi-fold doors etc, to address the applicable standards listed in this specification.

B. Performance Test Report

Submit reports including photographs of the performance test for the window wall, aluminum window, bi-fold doors, etc.

C. Planning of Inspection of Component

Submit preliminary schedule of material specification and procedure for inspection of the quality of components of the various systems during the fabrication in the factory.

D. Report on Inspection of Components

Report on the results of the inspection of the components in accordance with the "Planning of inspection of components" mentioned in above paragraph K.

E. Testing and Sampling procedures

1. During the process of finishes, testing shall be performed on specimens of equal metal thickness pretreated and finished along with the production metal. Specimens should exhibit a test area of at least 75mm by 300mm to permit instrument readings. In addition to running in-process tests to assure high quality production, additional finished extrusions or panels are to be submitted to the coating manufacturer's laboratory for extended exposure testing.
2. All test samples shall be properly identified with date, batch number, and shift indicated.

3. The following tests shall be conducted at least once per production shift and submitted to the Architect when requested:
 - a. Dry film thickness (evaluated with a Permascope, Isoscope, or ermatron instrument)
 - b. Film Hardness
 - c. Dry crosshatch adhesion
 - d. Boiling Water adhesion test
 - e. Gloss measurement
 - f. Colour examination against standard
 - g. General pinholes and other surface imperfections
 - h. Cure

Testing reports shall be certified by a testing agency, manufacturer and the contractor.

4. Procedures
 - a. Dry Film Hardness
The coating shall have a hardness of H minimum when tested with "Eagle Turquoise Pencil".
 - b. Film adhesion
Coating systems shall withstand the following adhesion test:
 - 1) Dry
Make ten parallel cuts 1.25mm apart through film and 10 more cuts 90 and crossing first 10 cuts. Apply Scotch Transparent #710 Tape, 18mm wide, over area of cuts, pressing down firmly against coating. Pull tape off sharply.
 - 2) Wet
Make 10 parallel cross cuts as above. Immerse samples in boiling water for 5 minutes. Removes sample, dry, cool and tape test the crosshatched area as above.
 - c. Gloss measurement
Measure gloss at various locations on painted metal with a 60 Glossmeter.
 - d. Colour Uniformity
Check random samples of painted production metal under a uniform light source, such as natural North daylight against standard panels accepted by the Architect.
 - e. Test for cure of coating using 100 double rubs with several thickness of cheesecloth wet with MEH solvent. Slight dulling of the film is considered normal, but softening shall not be permitted.

F. Window, Louvre & Door Field Water Testing

100% of the constructed Window, Louvre and Door all fully installed including associated waterproofing shall be physically tested in accordance with the requirements of AAMA 501.2-83, for confirmation of water tightness. Method of testing shall be as follows or approved by the Architect

1. Test for Static Water Penetration

No water infiltration under static pressure shall occur when the window is tested in accordance with ASTM E 331-86 at a differential static pressure of 3.8kN/m² or 20.8 lbs/sq. inch for 15 minutes. Using a water spray discharge rate of 200 liters of water per hour per square meter (five gallon of water per hour per square ft.).

2. Test for Dynamic Water Penetration

In accordance with NAAMM T-1-68T or AAMA standard TM-1-76, method of test for metal window walls for water penetration using dynamic pressure, at design pressure specified earlier.

3. Field Water Tests

Field water tests in accordance with NAAMM FC-1-69 shall be performed on completed portions. All remedial actions are to be at no additional costs to the Employer. (Equivalent AAMA FC-1-76 Specification may be accepted subject to the Architect's approval).

1.12 WARRANTY

- A. This part shall be read in conjunction with Clause 8.04 of Specification Preliminaries for the list of material/ system requiring joint written guarantees.
- B. Furnish a written guarantee to Employer warranting all works in connection with organic coating system to be free from defects in materials and workmanship for a period of fifteen (15) years from the end of Defects Liability Period or the date when the Architect issues the Certificate of Completion of Making Good Defects, whichever is the later, and to correct promptly any defects without additional cost to Employer. The following are considered defects without being limited thereto.
1. Peeling
 2. Cracking
 3. Checking
 4. Blistering
 5. Chalking in excess of #8 chalk rating when measured in accordance with ASTM D 659-44 (1965).
 6. Fading or colour change in excess of 5 NBS unit when calculated from measurement on a spectrophotometer or colourmeter capable of colour measurement by reflectance readings or accordance with ASTM D 2244-68.
- C. A written warranty shall be provided (in any event to be backed by a guarantee from the parent company or companies) agreeing to repair or replace defective materials and workmanship during the warranty period.
- D. In case the design, engineering or supply of any part of the works are to be done by a manufacturer other than the main contractor, the warranty shall be backed by a guarantee from the manufacturer(s) for repairing or replacing defective materials and workmanship during the warranty period. Defective materials and workmanship include, but are not limited to:
1. Any abnormal deterioration, deformation, clippings, aging, rusting, corrosion or weathering of the work.
 2. Water leakage under conditions equivalent to, or less severe than, those specified.
 3. Air leakage exceeding specified limits.
 4. Failure of operating parts to function normally.
 5. Structural failure due to pressures and forces up to specified limits.
 6. Sealant (including structural silicone) loss of adhesion, loss of cohesion, cracking or discoloration.
 7. Disengagement of gaskets or weather-strips.
 8. Deterioration or discoloration of aluminum and cladding finish in excess of normal weathering and aging.

9. Glass breakage due to pressures up to specified values, thermal stress, defective glass or damaged (prior to or during construction) glass, and resultant damage and breakage caused by falling glass.
 10. Spontaneous breakage of tempered glass – Submit a joint warranty with the manufacturer with performance pledge for not more than 1:1000 pieces (0.1%) breakage rate of the glass with heat soak test.
 11. Delamination or discoloration of laminated glass or panels.
 12. Loss of effective glass bite due to shifting of glass.
 13. Loss of effective glass bearing on setting blocks due to shifting of glass and/or blocks.
 14. Collapse of insulation.
 15. Failure of work to fulfill other specified performance requirement.
 16. Cracking, peeling or discoloration of glass reflective coating.
 17. Defects in structural anchor, stainless steel finishes and aluminum anodizing and powder coating / PVF2 coating.
 18. Colour and finish on aluminum which cracks, peels, fades in excess of specified limits or chalks in excess of specified limits.
- E. The warranty does not include damage caused by vandalism, or natural conditions exceeding the performance requirements. However, the guarantee does include failures or defects for which the causes cannot be determined. The warranty and its enforcement shall not deprive the Employer of other action, right or remedy available to him.
- F. The warranty for the works shall be **ten (10) years** from the certified date shown on the Certificate of Substantial Completion of the Main Contract issued by Architect. The system warranty includes all materials, plant and labour. All costs and expenses arising out of or in connection with the obligation herein shall be solely borne by the main contractor.
- G. Review and inspect the work of other nominated sub-contractors and shall report in writing any discrepancies or problems noted which could result in a warranty coverage conflict should any failures or problems occur. In the event of any disagreements between main contractor and nominated sub-contractors relative to the system performance and warranty coverage, the final decision concerning which party shall be responsible shall be made by the Employer, Architect and/or other consultants.
- H. Failure to provide special warranties that meet the above stated requirements may lead to disapproval of the proposed glass and materials and the main contractor shall re-submit glass and materials from alternative services that are able to provide special warranties that meet such requirements.
- I. All guarantee is to include for the cost of removal and replacement of all defective work at no extra cost to the Employer at such time as designated by the Employer during the warranty period.
- J. The wording of the guarantee shall be to the approval of the Architect and the Quantity Surveyor.

1.13 SPARE PARTS

This part shall be read in conjunction with Clause 8.05 of Specification Preliminaries for the list of spares and spare parts.

PART 2 PRODUCT

2.01 APPROVED MANUFACTURER

(NOT USED)

2.02 MATERIALS / SYSTEMS

A. General

1. Materials and components used shall be as specified or shall be suitable equivalents as accepted by the Architect.
2. Materials not listed shall be of the best quality and suitable for the purpose and as accepted by the Architect.
3. All materials shall be free from any defects that may impair the strength, functioning, or appearance of the works or adjacent construction.
4. Testing by independent testing laboratory or review of data by the Architect shall not mitigate the Contractor's responsibility for performance of the aforementioned systems, nor relieve the Contractor of his responsibility to verify for himself that the work conforms to the intent of the Contract Documents.
5. The thickness and type of glass shown on drawings are indicative only and its capability of withstanding all required wind loading, horizontal imposed loads, impact load as referred to in Table 3 of Building (Construction) Regulation, and other static and dynamic forces under all statutory regulations shall be the sole responsibility of the Contractor who should bear all costs for rectification when it is necessary;
6. Sizes, thickness, spacing, etc. of the constituent members and structural frame / hangers, etc. of various systems illustrated on drawings are indicative only. Verify, design and construct the various systems in accordance with the drawings and comply with this specification. Ensure the capability of their design systems in withstanding all the required wind loadings, horizontal imposed loads as referred to in Table 3 of the Building (Construction) Regulation, and other static / dynamic forces under all statutory regulations.

B. Aluminum

1. Production and Standard of aluminum
 - a. Names and particulars (including size of factory, quantity & type of production lines, job reference and etc.) of the manufacturing company for aluminum section and the fabricating factory (if different from the aluminum manufacturing company) shall be submitted to the Architect for review.
 - b. The factories should have ISO quality standard in the management of the factory, the production of aluminum and the quality control of its product. Relevant latest ISO certificate(s) should be submitted to Architect for verification before the production of the materials. If the factories fail to produce the required certificates, the materials produced by the factories would not be accepted and the contractor would be liable for the delay of works due to the late delivery of materials.
 - c. All ingots must be from Australia, Europe or North America. All sections may be manufactured in Southeast Asia or China and shall be formed true to detail and free from defect impairing appearance, strength or durability. Strength standard should conform to BS EN 515:1993, BS EN 485, BS EN 573, BS EN 755 and BS EN 12020.
 - d. All aluminum sections and capping shall be cut into size, dimensions & shape in accordance with the approved shop drawings before apply coating / finishes.

- e. Extrusion alloy and temper shall satisfy minimum requirement of B.S. H9TE Alloy, JIS 6063-T5, 6063-T6 or 6061-T6. Provide 3mm minimum nominal thickness for structural members and 1.5 mm minimum for non-load-bearing / non-structural part unless otherwise stated on the drawings and contract.
- f. Sheet alloy and temper shall satisfy minimum requirement of B.S. Alloy N3 or N4, JIS 1100, 1200, 3003-H14 or 5005-H14. Provide 3mm minimum nominal thickness for structural section.
- g. Tolerances as listed in "Aluminium Standards and Data" are applicable to finished, fabricated and assembled materials. Maintain stricter tolerances where required for proper fit of components, including gaskets and glass.
- h. All aluminum products shall be manufactured by the same factory.
- i. Fabrication and assembly shall be executed under factory conditions. Components or parts that are not required to be shop-assembled shall be shop-fitted and marked before disassembly, to ensure proper assembly at site.
- j. Materials, components and systems incorporated in the work shall be in compliance with the standards and procedures of the appropriate manufacturers.
- k. After acceptance of the fabrication factory and the shop drawings, arrange a factory visit for Architect and Client's representatives to the factory during the bulk production of aluminum sections / units at the cost of Contractor including traveling & accommodation expenses. The visit would aim at inspecting the quality control of the factory; the original certificate of the factory presented for factory review would be required for verification.
- l. Cladding corners and non-extruded corners to be v-cut joint unless otherwise stated in the drawings.
- m. Aluminium shall be finished in accordance with the standards and procedures developed and recommended by the manufacturer and as specified herein.
- n. At the option of the Architect, random production parts will be inspected in natural light. The method of inspection is at the discretion of the Architect, except that it will be by visual comparison within approved colour range standards.
- o. The Architect shall have final authority to accept or reject any or all material not meeting these finishing standards or other requirements of the drawings and specifications.
- p. Complete certified inspection records for quality of finish and complete finish process records shall be maintained. Such records will be made available to the Architect upon request.
- q. Material shall not be shipped, delivered or supplied when the finish of such material:
 - 1) Has not been inspected and tested in the manner and by the means prescribed above.
 - 2) Does not meet all the specifications for the finishes set forth in the manufacturer's instructions.
 - 3) Does not fall within the colour range approved by the Architect
 - 4) Has not been approved by the Architect
 - 5) Has not otherwise been processed in accordance with these instructions.

C. Aluminum Finishes

1. Clear Anodic Coating

- a. Clear anodic finish (natural) should conform to AAMA 607.1-1977 except those specified otherwise.
- b. All aluminum which will receive this finish shall be 6063-T5, 3003 or 3004 alloy as selected by the Architect.
- c. Provide Architect for selection a series of samples using both alloys and several finish variations.
- d. After selection of finish, prepare sufficient sets of two (2) samples each which shall define the colour range and submit them for Architect's review
- e. All samples shall be identified and have a full laboratory report attached describing full details of finishing process.
- f. Prior to the anodic coating process all material shall receive a caustic-etch treatment, in which a minimum of 25 micron (um) max. 75 micron surface material shall be removed from all visible surfaces.
- g. Anodic coating thickness shall be a minimum of 20 micron (um) and 25 micron max.; minimum coating weight shall be 4.4mg/sq. cm.
- h. Inspection after Finishing.
 - 1) Samples from production lots of finished material shall be tested in accordance with the following.

ASTM B137	-	Oxide Coating Weight
ASTM B244	-	Oxide Coating Thickness
JIS HB680	-	Oxide Coating Sealing
ASTM B136-72	-	Stain Resistance Test
ASTM B457	-	AZTAC, Test
ISO 3210	-	Acid Dissolution Test
JIS H8683	-	Oxide Coating Sealing Quality Test

Submit the result of the above tests to the Architect for his review and satisfaction.
 - i. Maintain complete finish process records including certified inspection records for quality of finish. Provide copies of records to the Architect upon request.
 - j. Repair of damage to finished surfaces by mechanical means (other than those specified). Repair of finishes by painting even in factory is strictly forbidden unless authorized in writing by the Architect.
 - k. Finished surfaces shall be free from mechanical imperfections such as scratches, scrapes and dents and free from finish imperfections such as spots, stains and streaks.
 - l. Aluminium finish at where structural silicone to be received
 - 1) Mill finish is not acceptable at structural silicone bonding surfaces.
 - 2) Aluminium surface to which structural silicone will be adhered shall have anodized or PVF2 finish.

2. Organic Coating – Fluorocarbons Coating (PVF2 Coating)

- a. All aluminum works to receive PVF2 coating (3 coats) unless otherwise specified in the drawings and tender documents.
- b. All products should have license from Pennwalt Corporation.

- c. After selection of colour by the Architect, prepare sufficient sets of samples each which shall define the colour and gloss range and submit them for Architect's review.
- d. All samples shall be identified and have a full laboratory report attached certifying the process and quality of the finishes.
- e. All aluminum sections shall be cut into size, dimension & shape in accordance with the approved / endorsed shop drawings before apply fluorocarbon coatings finish.
- f. The finishing system, including materials and application, shall conform to the requirements and recommendations of the paint manufacturer and the following :

1) General

- i Painted aluminum finish shall be factory oven cured three coat (minimum) finish based on Kynar 500 fluoropolymer resin (Polyvinylidene Fluoride-PVF2) as manufactured by Penwalt Corporation and supplied by one of their licensed formulators. The licensed formulator shall identify the finish meeting the minimal requirements for formulation as set forth in the licensing agreement.
- ii Application of the finish shall be performed under specifications issued by the licensed formulator and by an applicator specifically approved by one (or more) of the formulators. Applicator shall provide written notification of approval by a formulator prior to application of the finish.
- iii The formulation shall have at least 70% Kynar 500 resin of the residual solids.
- iv Name of the licensed Applicator shall be submitted to the Architect for review and acceptance.

2) Metal Preparation

Cleaning and pre-treatment process shall meet ASTM D 1730-67, Type B, Method 5 or Method 7. Processing shall conform with ASTM B 449-67 (1972) Section 5.

- i Cleaning and pretreatment system which shall remove organic and inorganic surface soils, grease, oils, residual oxides, etc. shall be followed by a chemical conversion coating and shall consist of the following:

Alkali clean	Mid etching
Rinse	Tap water, overflowing
Acid Deoxidize Rinse	Tap water, overflowing
Conversion coating phosphate	Amorphous chrome
Rinse	Tap water, overflowing
Final Rinse	Acidulated
- ii The conversion coating thickness shall range between 50 - 100 kgm per square foot.
- iii The above process shall take place immediately prior to application of the finish with a minimum of time lag between the two. There shall be no handling or lengthy storage of the pretreated metal prior to painting.

3) Finishing System

The finishing system shall consist of a shop applied, oven baked top coat over a recommended inhibitive primer, separately baked following paint manufacturer's instructions. Application should be performed by conventional air or electrostatic spray or reverse roller coating where

applicable, and thermocured separately in bake cycle of 15-20 minutes at 230°C. The finishing system shall consist of the following:-

Prime Coat - Apply a thin uniform film of primer at a dry film thickness of 0.3 ± 0.1 mil.

Top Coat - Apply top coat to obtain a fully opaque dry film of 1.0 mil (25u) minimum thickness and should not more than 75 micron. Gloss range shall be 25-35. Gloss measurements shall be made with a photovolt meter, at a 60 degree angle - in accordance with ASTM Standard Method D 523-67.

Colour final finish selection shall be made by the architect from samples prepared by Contractor.

3. Organic Coating – Polyester Powder Coating

- a. The polyester powder coating shall be baked applied by an approved applicator of the product under specifications issued by the manufacturer.
- b. The thickness of the coating should range from 40(um) to 120(um)
- c. The coating system shall be applied to mill finish aluminum which is properly cleaned and pretreated. The pretreatment shall meet ASTM D 1730. Type B. Method 5. Processing shall conform with ASTM B 449. Section 5.
- d. Baked polyester finishes shall be applied with one-coat bake systems. The coating shall have a minimum dry film thickness of 0.8 mils. (For more severe weathering applications consider alternative paints that include a primer coat). Recessed areas such as inside corners and channels are to be covered to the extent possible.

4. Organic Coating – Lilicatight Enamel

- a. Lilicatight Enamel shall be applied by an approved applicator of the product under specification issued by manufacturer.
- b. The enamel system shall be applied on surface treated (by acidic wash) galvanised mild steel surface in 2 coats (0.3 to 0.4 kg/M²) evenly by roller, brush or spray.

D. Glass

Production and Standard of Glass

1. Single source responsibility of glass

To ensure consistent quality of appearance and performance, and to comply with warranty requirements, provide materials produced by a single manufacturer or fabricator for each kind and condition of glass indicated and composed of primary glass obtained from a single source for each type and class required. All glass processes of fabrication including heat toughening, application of thermal coatings, ceramic enameled coatings and any process of lamination shall be manufactured by and fabricated into insulating units by a single source with the ability to comply with and assume all specified warranty provisions.

2. Read in conjunction with ASD General Specification edition 2012 section 20 for the standards and requirement of glazing in general. Where tempered glass is referred to in specifications, tender documents and drawings, it is to be defined as Fully Tempered Glass in ASD General Specification.

3. The glass to use shall be follows:

Glass shall be xinyi or approved equivalent in PRC origin

The acceptance on equivalent alternative is at the Architect's sole discretion and the Architect's decision shall be final.

4. All glass to be of clear glass, except where otherwise stated in the Window Schedule.

Glass color and appearance of the glass shall be selected by the Architect, and the Sub- Contractor shall submit the glass sample with performance data for Architect approval.

5. Heat soak test

100% Heat Soak Test is required for all tempered glass to be carried out in accordance with PNAP APP37 and BS EN 14179.

Prior to heat soaking, submit job references, a quality supervision plan, sample record forms for the test, location of fabrication, description of processing to be carried out, a written report to demonstrate oven and thermocouple calibration and temperature tolerances by an independent glass specialist, and other documents to meet the requirements in APP37, ASD General Specification and BS EN 14179.

The Architect or Employer's representative shall be given the opportunity of visiting the glass manufacturer's premises during fabrication and/or processing.

Prior to delivery to site for installation, provide full set of detailed records of heat soaking for each batch, with endorsement by the manufacturer and demonstrate with documentary evidence endorsed by an independent glass specialist that the glass has been heat soaked in accordance with APP37, BS EN 14179 and quality supervision plan. Such evidence to include, as a minimum, the following:

- a. Source of supply and evidence of batching;
- b. Dates and records of toughening and heat soaking of each batch of glass;
- c. Certification that the glass meets the performance requirements of the Specification;
- d. Records to include details of all units that failed during the heat soak test; and
- e. Other requirements in APP37, ASD General Specification and BS EN 14179.

6. Insulating glass shall not experience fogging, wetting or staining within the sealed space, space corrosion, spacer migration, adhesive or cohesive failure of primary or secondary edge seal. Insulating glass shall not experience decrease in the air space dimension due to chemical reaction of desiccant with entrapped air.

7. Tempered glass shall be subject to quality control measures with heat soaking to minimize inclusions which could result in spontaneous breakage. Such inclusions are defined as a material defect by this specification. Installed tempered glass which experiences spontaneous breakage shall be replaced under the warranty provisions.

8. All glass shall be free of visible scratches and digs and pinholes with the following criteria:

- a. pinholes with diameter exceeding 1mm are not acceptable, and are only permitted within 75mm of an edge;
- b. clusters of pinholes are not acceptable;
- c. scratches shall not have a length exceeding 75mm and are permitted only within 75mm of an edge;

9. Low-Emissivity (Low-E) Coated Tempered Insulated Glass Units

Location: All windows and glass doors

The unit shall be tempered glass with a clear pvb interlayer and a low-emissivity coating applied on the second surface. Properties of the glass shall be as follows or otherwise approved by the Architect:

Minimum Glass Thickness: Refer to window schedule.

Minimum glazing performance:

Visible Light(%)			Solar Energy(%)		U-value(W/m ² ·K)		Shading Coefficient	Solar Heat Gain Coefficient
Transmittance	Reflectance		Transmittance	Reflectance	Winter night	Summer day		
	Out	In						
60			11		12		29	30

Submit 2 specimens of glass unit 300x300mm with certificates from an independent HOKLAS laboratory verifying the performance of the specimen will meet the above performance.

10. Laminated Tempered Glass

Laminated glass shall consist of two layers of heat strengthened glass with equal thickness and an interlayer with 2mm nominal thickness. Glass lamination shall not delaminate, stain or discolour.

Location: Glass canopies, internal glass partition walls

Minimum Glass Thickness: 12 mm +2mm+12mm

11. The Contractor shall be responsible for all inspection and tests required under prevailing statutory requirements, including PNAP and conditions in Buildings Department approval letters. The Contractor shall submit all relevant certificates and documents in complying such requirements.

2.03 ANCILLARY MATERIALS

A. Sealant Material

1. Products:

2. Non-structural seals (exposed and internal): DOW CORNING 795" non-structural sealant (or approved equivalent)

Location: As specified; Samples / control samples: 10m run (each type) /as specified.

3. Structural seals: DOW CORNING 795 or 983" (or approved equivalent)

Location: As specified; Samples / control samples: 10m run (each type) /as specified

4. Data sheets and samples may be submitted for consideration of other silicone sealants, which are subject to approval. Oil based sealants are not acceptable.

5. General Sealants Requirements:

- a. Locate and identify all sealants by product name on shop drawings.
- b. In using specified sealants or approved alternates, strictly observe the printed instructions and recommendations based on tests of the sealant manufacturer regarding joint size limitations, mixing, priming, and application. Unless printed instructions advice to the contrary, do not apply sealants when substrates are wet or when the temperature is below 5°C.
- c. Sealant back-up materials shall be polyethylene foam, urethane foam or extruded silicone as recommended by sealant manufacturer.
- d. All sealants shall be neatly applied with taped edges and then tooled, as a separate operation, after application.
- e. Field application of sealants shall not be unduly delayed. Sealant work shall be performed at not more than a two to three floor interval behind the installation of panels and/or curtain wall units to prevent contamination of sealant contact surfaces and the accumulation of deleterious water infiltration within the wall cavity.
- f. The colour of sealant shall be selected by the Architect.
- g. Alternative glazing techniques may be proposed to suit variations in frame and thickness of reflective glass.
- h. The Contractor must ensure that the silicone proposed is compatible with the type of glass and cladding used for the project. The Contractor shall

obtain from the glass manufacturer a Guarantee stating his acceptance of glazing details, conditions and glazing material. Endorsement of any sealant of silicone or glass product by the Architect shall not relieve the Contractor of such requirement.

- i. Sealants used for weather seals shall not experience adhesive or cohesive failure. Sealants shall withstand movements up to the limits prescribed by the manufacturer. Exposed sealant surface shall not crack or bubble. Sealants and primers shall not stain adjacent materials.
- j. The silicone for structural glazing should have minimum bond strength to the metal of 480KPa (70 PSI). The sealant, when properly applied shall transfer the wind load from the glass to the structural members. This structural glazing application should conform to the design stress on the silicone joint configuration not exceeding 20 PSI representing a 3 to 1 safety factor recommended by sealant manufacturer. Evidence must be furnished to the Architect that the sealant manufacturers have reviewed their shop drawings and accepted the details for silicone performance. Details of the design width of sealant shall be submitted with the tender.
- k. Tensile or shear stress in sealants should not exceed the figures as recommended by sealant manufacturers.
- l. Consistent and periodic adhesion testing, especially for structural glazing, shall be carried out during the fabrication period to identify potential problems before materials are put into place on the building.
- m. Records for sealant batch number and date of manufacturer, as well as units to which the sealant is applied shall be maintained and made available to the Architect upon request.
- n. Testing and Passing Criteria:

Testing and Passing Criteria:

Physical Property	Test Method	Performance Requirements
Hardness, Durometer A	ASTM D2240	60 ± 10
Tensile strength	ASTM D412	2000PSI, min.
Elongation at break	ASTM D412	300% minimum
Brittleness temperature	ASTM D746	-40F
Tear Resistance	ASTM D-624	150 lbs./linear inch minimum
Flame Resistance	ASTM C-542	Must not propagate flame
Resistance to heat, aging, change in original properties after 70 hrs at 100 C (212 F)	ASTM D573	+/- 10 points, maximum
Hardness Elongation		-40% maximum
Tensile Strength		-15% maximum
Resistance to oil aging change in volume after 70 hours immersion in ASTM oil #3 at 212F.	ASTM D-471	+80%, maximum

Resistance to ozone Condition after Exposure to 100 pphm ozone in air for 5000 hrs at 100 F. (Sample under 200% Strain)	ASTM D1149	No cracks
Resistance to permanent set - Compression set after 22 hrs at 158 F.	ASTM D395 (Method B)	30% maximum
Resistance to water change in weight after 96 hrs. immersion at 212 F.	ASTM D-471	+10% maximum

Adhesive - Shall be neoprene adhesive as recommended by the neoprene manufacturer. Lap and adhere, or seal, all splices as recommended by the neoprene manufacturer.

B. Gaskets And Setting Blocks

1. Sponge gaskets shall be extruded black neoprene with a hardness of 40 +/- 5 durometer Shore A and conforming to ASTM C509-70. Design sponge gaskets to provide 20% to 35% compression. Cellular (closed cell) neoprene shall conform to ASTM D-1056 SCE 41.
2. Dense gaskets shall be black extrusions with a Shore A hardness of 75 +/- 5 for hollow profiles and 60 +/- 5 for solid profiles, and conforming to ASTM C 864-79. Outdoor gaskets shall be neoprene. Indoor gaskets shall be neoprene or EPDM.
3. Injection-mould all corners of gaskets where compatible with installation procedures.
4. Interior and exterior gasket profiles shall be designed to produce a glass edge pressure of not less than 182N (4 lb), nor more than 450N (10 lb) per 25.4mm.
5. Gaskets used in contact with structural silicone glazing materials shall be of extruded silicone, acceptable to the sealant manufacturer.
6. Due to the interaction of the exterior weather seals with the exterior gasket, the exterior gasket must be compatible with weather seal sealant.
7. As an alternate to extruded silicone gaskets for the gaskets used at structural silicone joints, "Norton V-2100" tape may be used, if proper detailing is provided. (Compatible neoprene gaskets should be considered only as the last alternative).
8. Setting blocks, where required, shall be dense extruded neoprene or silicone setting blocks are required in areas of structural glazing with a hardness of 85 +/- 5 durometer Shore A, a minimum length as required by the glass manufacturer and a minimum width corresponding to the glass thickness and retaining member. Locate setting blocks at quarter points.
9. Shims used in conjunction with setting blocks must be of the same material, hardness, length and width as the setting blocks.
10. Setting blocks and chair shall be secured against migration.
11. Locate side blocks where required within the upper half of each jamb for each light. Blocks shall be 55 +/- 5 durometer Shore A dense neoprene or silicone. Install block with 3mm clearance between block and bearing surface. Block shall be sufficient length to prevent point loading on the glass.

12. Side blocks are not required where an individual glass light is continuously sealed with silicone at two or more edges.
13. The tender submission shall include the proposed material and specifications for all gaskets, weather-strips, setting blocks, spacers, glazing channel cleaner, glazing channel primer/sealer, etc. to be reviewed by the Architect.

C. Protective Materials For Metals

1. Protective materials for steel shall be one of the following:
2. Zinc coating (hot dip galvanizing) for shapes plates, bar and strip shall conform to ASTM A-123.
3. Zinc coating (hot dip galvanizing) for fasteners and hardware shall conform to ASTM A-153.
4. Zinc coating (hot dip galvanizing) for sheet shall conform to ASTM A-525, coating weight 0.61 kg/m. (2oz./sp.ft.).
5. All steel except anchoring fasteners shall be galvanized and receive two coats of Zinc Chromate primer; one coat applied in shop and one coat in field to include any damage of shop applied coating.
6. Zinc Chromate primer shall conform to FS TT-P-645.
7. At aluminium surfaces in contact with masonry, concrete or steel, prime paint with zinc chromate or bituminous paint. Before applying the primer, the aluminium surface shall be washed with white spirit to remove dirt and grease and any surface corrosion shall be removed by brushing with wire brush.
8. After field welding, remove weld slag and touch up finished surfaces with an approved epoxy based zinc rich primer coating. Before applying touch-up coating, the steel surface shall be thoroughly cleaned of all dirt, dust, grease, stains, rust, mortar splatter, loose material or other foreign matter.
9. Provide minimum dry film thickness of 0.025mm for each coat of zinc chromate and 0.75mm for bituminous paint. Zinc chromate primer shall conform to GSA specification TT-P-645.
10. Bituminous paint shall conform to FS TT-C-494.
11. per 25mm PVC coated open cell urethane foam, compressed 30 to 50 percent.

D. Concealed Flashing

1. Acceptable materials are stainless steel sheet, neoprene sheet and anodised or painted aluminium sheet. Minimum thickness is 1.5mm for neoprene and 1mm for aluminium or stainless steel. Aluminium is required to have a bituminous coating against dissimilar materials.
2. Provide sealed lap joints, end dams and transitions to gutters.
3. Sheet neoprene shall be 1.5mm thick with joints lapped and sealed 150mm minimum.

E. Anchors In Concrete And Masonry

1. Anchors embedded in concrete and masonry shall be hot dip galvanized hot rolled steel, or hot dip galvanized cold-formed steel.
2. Strength of embedded anchors shall be developed by integral projections of the steel anchor, or be welded deformed bars or headed studs.
3. For masonry, through bolts are acceptable provided that bearing plates are used at both masonry surfaces.
4. Expansion bolts set into drilled holes are acceptable at concrete but not at masonry. Self-drilling, self-threading screws are not acceptable.

5. Powder-actuated anchors bolts are not acceptable.
6. Epoxy/chemical anchor bolts are not acceptable.
7. Welds to anchors shall be touched up with an approved epoxy based zinc-rich primer.

F. Fasteners

1. Fastener requirements listed herein are applicable to screws, bolts, nuts, washers, rivets and pins.
2. Fasteners in contact with cementitious material shall be stainless steel type 304.
3. Fasteners outboard of or within a glazing pocket, gutter, flashed-cavity or other potentially wet location (after completion of construction) shall be stainless steel type 316. Fasteners inboard of potentially wet locations shall be stainless steel type 304, cadmium plated carbon steel or zinc plated carbon steel.
4. Provide lock washer or other locking device at all bolted connections.
5. Powder-actuated fasteners are not acceptable.

2.04 FABRICATION

(NOT USED)

2.05 FINISHES/FINISHING TREATMENTS

(NOT USED)

2.06 STORAGE

(NOT USED)

2.07 WORKMANSHIP

A. Construction

Aluminium window, louvre and door to BS4873 : Part 2 in accordance with approved Shop Drawings.

B. Fixing

Fix aluminium window, louvre and door as follows:

1. Avoid distortion during handling and storage. Keep opening lights closed.
2. Position, plumb, level and square.
3. Form or cut pockets in heads, jambs and cills etc. to receive fixing lugs, build in lugs and make good and screw to frames or plug and screw frames using packing pieces where necessary. Do not distort frames when tightening fixings.
4. Bed around frames in butyl mastic bedding compound (waterproofed cement sand grout 1:1) leaving no gaps. The use of timber for packing around windows is not permitted.

PART 3 EXECUTION

3.01 EXAMINATION / INSPECTION

(NOT USED)

3.02 PREPARATION

(NOT USED)

3.03 INSTALLATION / APPLICATION

A. General Construction

1. The Architect's drawings shall be considered essentially schematic, except for the Profiles of exposed surfaces and the module of Window, Louvre and Door which shall be as indicated.
2. The Architect's specification describe minimum requirements with respect to material, fabrication, assembly and installation of the items specified and shall be used only as a guide.
3. If, in the opinion of the Contractor, a change of profiles or module is required in order to meet the specifications, he shall consult with the Architect for a review of the condition.
4. The method of assembly, reinforcing and anchorage of the Window, Louvre and Door System, where indicated, is schematic. Locations and method of providing same shall be the Contractor's responsibility, who shall design the assembly, reinforcing and anchorage to suit each specified condition in an acceptable manner complying with the requirements specified herein.
5. Materials, methods of fabrication, assembly, erection, fastening, supporting and bracing shall be in accordance with the approved shop drawings.
6. The Contractor shall not use any material, equipment or practices that may adversely affect the function, appearance and durability of the completed Window, Louvre and Door and related construction. The work shall be accomplished in compliance with the specified criteria without buckling, opening of joints, cracking of glass, leakage, noises, or other harmful effects.
7. The Window, Louvre and Door shall be installed according to the approved Shop Drawings, set plumb, square level at proper elevation and located in proper alignment with all adjoining work. All members shall be securely anchored in accordance with approved details. All anodized aluminum surfaces which are in contact with masonry work shall receive a protective bituminous coating or zinc chromate paint. All aluminum in contact with other metal shall be protected by an intervening impervious nonferrous material.
8. All exposed exterior metal components of the Window, Louvre and Door shall be designed, fabricated and installed in such a manner that they produce a visual flatness when viewed from any angle.
9. All steel fasteners and accessories shall be galvanized (hot dip galvanizing after fabrication) and shall receive anti-corrosive paint.
10. All other steel shall be galvanized and shall receive two coats of zinc chromate primer; one coat applied in the shop and one coat on site.
11. All site welds on steel shall receive two site coats of primer, using the same primer and surface preparation as those used in the shop. In case of galvanized surface, such welds shall be treated with an approved site cold galvanizing process (e.g. Ferraloy, Galvaloy).

12. Where the design requires moving metal parts, such parts shall be separated and coated or treated to ensure proper movement without noise, binding or metal to metal contact.
13. Isolate materials where necessary to prevent electrolysis between metal to metal contact.
14. Coat all metal surfaces where in contact with masonry, plaster or concrete surfaces with bituminous paint at a minimum coverage of 1.0 litres/sqm.
15. Use silicone sealant for joint subject to movement and in glazing.
16. Surfaces to receive sealant shall be properly prepared, cleaned, primed and excess sealant removed from finished surfaces.
17. Sealed joints shall be neatly trowled and surface smoothed. Follow the instruction of the sealant manufacturer.
18. Color of the exposed sealant shall be approved by the Architect.

B. Welding

1. Welding of steel shall be done in accordance with the relevant British Standard and/or the recommendations of the American Welding Society.
2. All welding shall be done with electrodes and/or methods recommended by the suppliers of the metals being welded. The type, size, and spacing of welds shall be as shown on the approved Shop Drawings. Welding materials and methods shall be such as not to cause distortion, discoloration, or result in any other adverse effect on the required profiles and finishes of the exposed Window, Louvre and Door.
3. Weld splatter and welding oxides on exposed surfaces shall be removed. All exposed welds shall be finished to match and blend with adjacent parent metal prior to application of the final finish.
4. No screws on outer surfaces of the sill members of the window frames shall be permitted

C. Window & Door Types

1. Openable Sash General
 - a. Mechanical joint construction is acceptable in lieu of welding. The connection joint's rigidity shall not be inferior to the welding construction.
 - b. The Contractor is requested to carry out structural tests on the sash framing according to the AAMA testing requirements, for Architect's approval
 - c. In addition, the openable sash shall be subject to operation tests to prove its durability of open/close operations and operating hardware's actuating mechanisms to meet the AAMA testing requirements, for Architect's approval.
 - d. Openable sash shall be provided at positions as shown on the drawings to keep the required ventilation.
 - e. Thermal insulation in accordance with DIN EN ISO 10077 part 1, $UD < 2,0 \text{ W/(m}^2\text{K)}$. Resistance to driving rain - class 9A in accordance with DIN EN 12208
 - f. Aluminum components shall be fabricated with joints accurately fitted and factory sealed and securely jointed to make tightly closed joinery.
 - g. Hardware for openable sash shall be designed to adequately perform the function for which it is intended and shall be securely attached to the window, and/or adjacent window.
 - h. The openable sash when in closed position shall remain watertight under all weather conditions and pass the water tightness tests as specified.

- i. The compressible neoprene weather-stripping shall be installed in one continuous piece, thereby, preventing separation at corners.
 - j. Top hung projecting out openable sash shall be fitted with:
 - k. Hinged, or parallel link arms;
 - l. Slide arms;
 - m. Offset type casement handle; and
 - n. Locking contact points shall be so spaced to assure even compression of weather-stripping over entire length and height of sash.
 - o. Hinges or parallel link arms shall be concealed stainless steel hardware.
 - p. All hardware and accessories shall be supplied by the Window Sub-Contractor, and when exposed, shall be of color and finish to match window members where possible. Numbers and strength of hardware shall comply with the wind load requirement. Proposal on the complete set of hardware to be used shall be submitted with the tender.
 - q. The detailed system of the openable sash must be proposed by the tenderer as shown on the drawings.
 - r. All hardware and fittings, including hinges, for the openable sash shall be stainless steel, heavy duty, from leading manufacturer's products such as ANDERBERG of U. S. A. or equivalent. No opening through frame for fixing the hardware.
 - s. Limiter is required for all openable window
2. Lift & Slide Door
- a. Aluminum components shall be fabricated with joints accurately fitted and factory sealed and securely jointed to make tightly closed joinery.
 - b. Vent/leaf profiles and roller carriers for vent/leaf weights of up to 250 kg must be used.
 - c. Thermal insulation in accordance with DIN EN ISO 10077 part 1, $UD < 2,0 \text{ W/(m}^2\text{K)}$. Resistance to driving rain - class 9A in accordance with DIN EN 12208
 - d. The bottom, replaceable tracks are made of stainless steel.
 - e. The frame connections of the profiles of the outer frame and vent/leaf are made with precision corner cleats.
 - f. The sliding vents/leaves are locked with:
 - g. multipoint locks and handles (used from inside)
 - h. multipoint locks and handles (used from inside and outside) as confirmed by the architect
 - i. The lift and slide vents/leaves are locked with lifting mechanisms and crank handles, used from inside
 - j. Toplights (fixed and turn/tilt) and/or sidelights must be manufactured with the same basic depth (120 mm as supported by structural calculations and as confirmed by the Architect) as the sliding and lift and slide units.
3. Louvres
- a. Louvers shall consist of extruded aluminum perimeter frames and blades. Frame corners and blade ends shall be welded or fastened with 300 series stainless steel screws. Provide concealed extruded aluminum stiffeners for blades, such that vector sum of blade deflections parallel to blade principal axes does not exceed 1/175 times span at design pressure.
 - b. Co-ordinate free area of louvers with mechanical requirements.

- c. Dummy louvers shall be closed by aluminum sheet with minimum 3.0mm nominal thickness, fastened and sealed to the indoor louver surfaces.
 - d. Finish for louvers and screens shall be the same as for other aluminum members.
 - e. The louvers shall be supplied in either modular or breakdown form and shall comprise:
 - f. External cill, side and top frames of channel section, hidden vertical support mullions with integral drainage channels, horizontal louver blades, including inner modules and blades for double bank and triple bank version which shall be separated from front external blades to ensure water separation.
 - g. All component parts shall be aerodynamically shaped and cold roll formed to ensure conformity of shape, or extruded where specified. The louvers shall incorporate a drainage system such that water collected within each panel is drained down via the mullion and internal drainage channels and down the louver face.
 - h. Birdguards, where required, shall be of equal material and be fitted to the rear of each panel and manufactured from expanded metal mesh. Blanking plates where required shall be of equal material.
4. Fixing Lugs
- Galvanized mild steel fixing lugs spaced at 300mm centres (maximum) for outer frames of each unit. Where specified fix lugs with rag-heads or "Ramset" fixing bolts.
5. Fastening Devices
- a. Aluminium alloy, stainless steel or nylon for all exposed fixings including screws, nuts, bolts, washers and rivets and match up with finish where possible.
 - b. Use stainless steel or steel with an anti- corrosion treatment for all concealed fastening devices.
6. Fittings
- All fittings including friction grips pivots, casement fasteners, spring catches, brackets, cement stays, slide arms, locks, slip bolts and handles of anodized aluminum to match the finishing or stainless steel, as specified, to retain the opening part rigidly in both the open and closed positions.
7. Contact Of Dissimilar Metals
- Avoid contact aluminum alloys with copper alloys, nickel, lead or steel. Cadomium plated or galvanized fittings can be used safely in contact with aluminum.
8. Drainage Of The Construction
- Rebates and profile grooves that may be penetrated by rain and condensation must drain to the outside. Visible drainage slots must be covered by caps.
9. Glazing
- a. Glazing shall be accomplished from the inside of building wherever possible.
 - b. The glazing system shall be designed to this end using continuous neoprene compression gaskets on both sides (preset gasket on one side of glazing pocket and roll-in gasket on the other side). A continuous "wet" seal shall be employed to insure a complete watertight.
 - c. Maintain a minimum glazing bite, edge clearance and surface clearance depending on the glass used as recommended by the glass manufacturer.
10. Glazing Beads

Glazing with approved glazing system and aluminum beads. Securely clip beads to the frame or use beads which are an integral part of the frame.

11. Glazing - Structural Silicone

The glass to be glazed with structural silicone sealant applied in strict accordance with the manufacturer's published instructions and shall be applied only by workman specifically trained or experienced in their use.

12. Sealant and Gasket Application

- a. Sealant and gasket shall be provided where shown on the drawings or required for a permanently weather tight installation. The sealing mechanism for each location and use shall be as indicated on drawings. In those locations where a mechanism is necessary but is not indicated, it shall be of type recommended by the Contractor and approved by the Architect.
- b. Protect all adjoining surfaces not to receive sealant against staining by masking and/or other methods.
- c. Joints and joint surfaces shall be clean, dry, and free of any material that may have an adverse effect on the bonding and/or seal of the sealant and gasket materials.
- d. Apply sealant and gasket under the conditions recommended by the manufacturer(s). Prime all surfaces to receive sealant and gasket unless recommended otherwise, use no sealant that has started to set in its container or a sealant that has exceeded the shelf life published by the manufacturer.
- e. Fill all joints continuously and completely with sealant, forming neat, uniform, concave beads. Finish the material flush with adjoining surfaces unless otherwise shown on the drawings. All sealant surfaces shall be tooled smooth.
- f. The sealant must be bonded to the adjacent structural components so that it will not come away from the bonding surfaces. PVC-U profiles must not come into contact with materials containing bitumen. DIN 18540 and the manufacturer's fabrication guidelines must be observed when sealing installation joints with elastic sealants.
- g. Tensile or shear stress in structural silicon sealant joint shall not exceed 1.4 kg/sqm or less if required by sealant.

13. Assembly And Anchorage

- a. Anchor component parts securely in place by bolting, welding or other permanent mechanical attachment systems that will comply with performance requirements and permit movements that are intended or necessary. Install slip pads between moving parts.
- b. Provide a separator at contact surface of dissimilar materials whenever there is a possibility of corrosive or electrolytic action.

14. Anti-Galvanic Action & Electrolytic Protection

- a. Isolate dissimilar metal surfaces to prevent galvanic action. Materials used for this purpose shall be non-porous such as neoprene, bituminous or zinc chromate paint.
- b. All steel parts shall receive a protective treatment commensurate with their respective functions. The treatment shall be one or more of those described above, and approved by the Architect.
- c. Aluminium surfaces in contact with mortar, concrete fireproofing, plaster, masonry, and absorptive materials shall be coated with an anti-galvanic moisture- barrier material or with bituminous paint of 1.0 litre/sq.m minimum coverage.

- d. Metal surfaces should be separated in such manner that metal does not move on metal. Where design requires moving metal parts, such parts shall be separated and coated or treated to ensure proper movement, without noise; or metal-to-metal contact. Materials used for this purpose shall be lubricating devices, sealants or gaskets

3.04 TOLERANCES

- A. Any parts of the Window, Louvre and Door, when completed, shall be within the following tolerances:
 1. Deviation from plumb, level or dimensioned angle must not exceed 3mm per 3.5mm of length of any member, or 6mm in any total run in any line.
 2. Deviation from the exact position on plan or elevation, including deviation from plumb, level or dimensioned angle, must not exceed 9mm total at any location.
 3. Change in deviation must not exceed 3mm for any 3.5mm run in any direction.

3.05 CLEANING

- A. Upon removal of all protection the Contractor shall clean the exposed surfaces of all work cleaned and left free of smears, scratches, and abrasions, to the satisfaction of the Architect. Any damages to finishes shall be required/replaced at no additional cost to the Owner.
- B. Select method of cleaning that will promote the achievement of uniform appearance and stabilised colours and textures for materials that weather or age with exposure. The method of cleaning should meet the recommendation of material manufacturers. In no case shall abrasive agents be used to clean any surface. Care shall be taken during cleaning to avoid scratching of the surface by grit particles. Remove and replace damaged or broken glass, regardless of how caused, with the specified material up to the date of acceptance of the work.
- C. Provide written verification that cleaning agents are compatible with finished aluminium, glass, glazing materials and sealants.
- D. Immediately before the time of final acceptance the Contractor shall demonstrate proper cleaning methods to the Employer's maintenance personnel. Prepare a 'Cleaning and Maintenance Manual' listing the types of cleaning components, cleaning methods, and the types of sealants and glazing materials to be used for repair and maintenance of the work.

3.06 PROTECTION

- A. Protection and Packaging of Materials
 1. Package and store materials in a manner that will prevent surface damage or contamination, distortion, breakage or structural weakening.
 2. Replace any materials damaged during manufacture, shipping, storage, or erection.
 3. Protect materials in place from contamination and damage.
 4. Submit a preliminary plan on the method for the protection of the surface of the aluminum window, glass canopy, aluminum features, claddings, and all other components in particular protection against typhoons, cement droppings, etc. during delivery prior to installation and during & after erection, with description as to when the protection can be removed.
 5. This plan shall be acceptable to the Architect.

6. The plan of protection shall include allowance for protecting all installed materials by encasing, covering up, etc. All proposed protective measures for the different components, accessories, glass, ironmongeries, etc. as against damages due to building operations (such as by covering up with adhesive tape, foam / timber casing, or plastic sheeting over the components upon installation to prevent from corrosion by cement slurry from building operations above or falling objects, etc.), adverse weather or other causes.
 7. The plan of protection shall include screening the various system components from weld splatter, spray-on fireproofing, concrete, alkaline masonry washes, paints, and other deleterious substances. Any such soiling shall be promptly and completely removed. The design of protective screening shall be such as to provide adequate ventilation of the space between the glass and the protective screen and not induce thermal stresses in the glass. In no case shall the protective screening be placed in contact with the glass.
- B. Wrap all window and door units and other associated materials in stout waterproof paper or polythene to protect against damp and scratching and do not make premature delivery to site. Apply a strippable coating or masking tape to all exposed aluminum or stainless steel surfaces.
 - C. Return to site on completion of building work to remove protective coating and leave clean. Lubricate and adjust moving components, and leave in perfect working order on completion.
 - D. The Contractor shall adequately protect all components and accessories from damage during shipments, storage at job site, erection and after completion of the work. At such time as may be directed, the Contractor shall remove all protective tapes or coating, thoroughly clean all anodized aluminum and glass surfaces with suitable cleaning agent, make final adjustments to all openable sash, etc. and hardware leaving all in first class working order.
 - E. All glazing and metal part when handover shall be free of digs and scratches.

END OF SECTION